

5V from Portable Wind Turbine

T.K. HAREENDRAN

This project deals with the design and development of a portable wind turbine unit, capable of generating electricity from the kinetic energy in the wind. The circuit requires a DC motor, fan blades or propeller, DC-DC boost converter and wind energy to produce a 5V DC output.



Fig. 1: Miniature DC motor



Fig. 2: Wind turbine blades/propeller



Fig. 3: DC-DC boost converter

Wind-power generation is a fairly simple process that uses an ordinary miniature DC motor to make a very simple wind turbine generator. A miniature DC motor, like RF-300FA-12350, is easily available in the market but can also be taken out from an old CD/DVD drive/player (refer Fig. 1).

A small propeller, or fan blades, can be mounted directly onto the motor shaft. Note that the electricity generated by the motor is DC, and hence no AC-DC conversion circuitry is required apart from a polarity guard diode. In fact, the polarity guard diode is also not required since the propeller rotates in only one direction (counter-clockwise) in this model, due to the unique design structure of the plastic wind turbine blades/propeller.

Compact and light-weight plastic fan blades/propellers are available. A generic three-leg plastic propeller (shown in Fig. 2) is used in this project.

Since a miniature DC motor is used as the rotary engine, only a

PARTS LIST

Semiconductors:	- 1N5817 Schottky diode
Capacitors:	- 47µF, 25V electrolytic
Miscellaneous:	- 2-pin connector
CON1	- DC-to-DC boost converter (PFM) module
BOARD1	- RF-300FA-12350 DC motor
M1	- Propeller (three-leg big plastic)

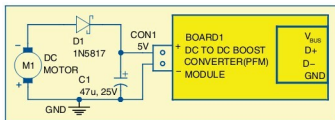


Fig. 4: Circuit diagram for deriving 5V from the portable wind turbine

small amount of DC supply (three to five volts maximum) is available at the output. So, a DC-DC boost converter circuit is required to provide a stable 5V DC output. Any home-made/ready-made DC-DC boost converter circuit can be used for this, however a ready-made DC-DC boost converter module based on pulse frequency modulation (PFM) technique with a standard USB output port, as shown in Fig. 3, is preferred.

If an input voltage of 0.9V to 5V DC is available, this module gives a stable 5V DC output through its USB connector, with conversion efficiency up to 96 per cent.

Circuit and working

As shown in Fig. 4, the circuit is built around a DC motor, Schottky diode 1N5817 and DC-DC boost converter module. Working of this small DC-DC boost converter module is based on pulse frequency modulation (PFM) technique.

A PFM converter is an alternative DC-DC power converter architecture that uses a variable frequency clock to drive power switches and transfer energy from input to output. Because the drive signal's frequency is directly controlled to regulate the output voltage, this architecture is referred to as PFM. DC-DC converter with constant-on-time or constant-off-time control is a typical example of this architecture.

In the circuit diagram, the negative terminal (black wire) of DC motor M1 is treated as the positive output terminal, connected to PFM

